



A Comparative Analysis of Machine Learning Algorithms for Heart Diseases Prediction

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Abstract—

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The term heart disease refers to the range of conditions that impacts heart and its blood vessels in a negative manner. This has led to increase mortality rate worldwide and a major health issue in recent years and continues to have increasing statistics in future. Modern day lifestyles and choices has made it increasingly common, not only in aged population but also in younger adults. New generation advanced medicines are fortunately a breakthrough in assisting and curing the disease, but conventional techniques are only optimal to provide on the spot single data reading and result, it fails to predict the future circumstances and possible emergency which may be caused due to underlying issue, or provide with early diagnosis to facilitate prompt medical assistance. This is where Machine learning (ML) comes into the picture, delivering the required and necessary early analysis of cardiovascular disease (CVD) through validating various features among thousands of data in a dataset. This study has developed predictive models and used Cleveland dataset with 11 different features namely age, sex, cholesterol, resting heart rate, exercise-induced angina and other health-related indicators that make up this dataset. There are four different predictive models that we will be using here to predict the health analysis of heart related disease, which are Logistic Regression, Random Forest, Support Vector Machines (SVM), K-Nearest Neighbor (KNN). All of the mentioned models are developed and used under Google colab

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platform. There are 4 different datasets consisting of Cleveland, Hungary, Switzerland and VA Long Beach which are preprocessed and cleaned and then being used and tested here with machine learning models to test its accuracy and other result aspects. The best result and performance among these four models tested across each of the datasets and having minimum spread (max-min) is of Support Vector Machine (SVM). with an F1 score of 0.7732. Provide a clear, well-structured, and informative summary of your research that highlights the essential aspects of the study. The abstract should briefly explain the purpose and significance of the research, outlining the problem addressed and the motivation behind the work. It should then describe the methodology or approach used, including key techniques, tools, datasets, or theoretical frameworks applied in the study. Following this, summarize the major findings or results, emphasizing the most important outcomes, observations, or contributions generated through the research process. Conclude the abstract by presenting the primary implications, conclusions, and potential applications of the work, as well as its relevance to the field of study.



The abstract must be written in a concise, objective, and reader-friendly manner so that it can stand alone as a complete overview of the paper. It should accurately reflect the core contribution and novelty of the research without including unnecessary background information or overly detailed explanations. Avoid the use of citations, references, figures, or tables in the abstract, and do not include undefined abbreviations or technical jargon that may limit readability for a broader academic audience.

The content should be logically organized in a single paragraph, maintaining coherence and clarity throughout. Ensure that the abstract captures the research context, problem statement, approach, key results, and final conclusions in a balanced manner. It should provide enough detail to help readers quickly understand the scope and value of the study while encouraging them to read the full paper. The recommended length is between 150 and 250 words; however, it may extend up to 500 words if necessary to clearly communicate the research objectives, methods, findings, and significance.

Keywords— List 4–6 relevant keywords separated by semicolons.

I. INTRODUCTION

Heart, one of the major organs in human body, consists of two upper atria (receiving chambers), two lower ventricles (pumping chambers), septum which separates the two sides of heart, valves and arteries. Additionally, the overall function primarily starts with the heart walls and electrical system of heart. The outer layer (epicardium), thick muscular layer (myocardium) responsible for contractions and the inner endocardium. Its rhythm is controlled by the electrical conduction system, which coordinates the heart's muscles to contract and pump blood at a normal paced rhythm allowing the blood to flow into the body and vice versa.

Any restriction, blockage, electrical misconduction or irregularities in heart can cause a serious medical emergency.

Cardiovascular disease (CVD) has been the leading cause of death globally, with increasing statistics every year with 18.5 million in 2023, 20.5 million in 2025 and projected to 35.6 million by 2050 by European Journal of Preventive Cardiology. Blockage in artery due to fat or calcium deposits, congenital disease,

cardiomyopathies, hypertension are the upmost reasons.

Further, unhealthy or sedentary lifestyle, smoking, substance abuse, poor diet, bad stress management also increases the risk of occupying these diseases.

Conventional techniques are accurate, reliable and are among the options for treating and evaluating patients, such as angiography, echocardiogram (ECG), stent insertion, statins, vasodilators and many more. But these are invasive procedures, resource-intensive and have varied effectiveness among different people.

Technological advancements have significantly impacted the prevention and treatments in a positive manner. Techniques such as gene editing (One-time in vivo CRISPR editing of PCSK9/ANGPTL3; potential "one-and-done" lipid therapy), injections such as Inclisiran (siRNA PCSK9i, sustained LDL-C reduction ~50%), SGLT2 inhibitors (Cardioprotective beyond glycaemic control), Lp(a)-targeting RNA therapies (80–95% Lp(a) reduction; targeting a previously "untreatable" genetic risk factor) have

promising solutions in reducing the major risk factors of CVD, by vast difference.

One of the important areas of curing and preventing CVD is early diagnosis, detection and accurate future health marker prediction, which somehow cannot be tackled using conventional techniques, as most of the patients or people are reluctant to participate in clinical trials, routine health checkups. Here Machine learning (ML) plays a big and important part, defining a new way to not only identify and diagnose CVD, but with low-cost and reasonable accuracy. ML techniques for diagnosis does not require multiple clinical trials, as different records of various patients with their features can be used ethically to predict and evaluate CVD with high accuracy.

It is expected that ML application will be globally used to increase doctor's work efficiency, generate economic benefits.

II. LITERATURE REVIEW

Summarize previous work relevant to your study. Present key contributions, methodologies, and findings from earlier research. Discuss limitations or gaps in existing studies that your work aims to fill. Group similar studies together for clarity, and cite appropriately. Highlight how your research builds upon or differs from prior work in the field. Explain the rationale behind your chosen approach and how it addresses identified gaps or limitations. Clearly state the unique contributions and potential impact of your study within the broader research context.

III. METHODOLOGY

The methodology followed in this work encompasses data collection, preprocessing, experimental setup, model selection, and performance evaluation. A complete description of each stage is provided below.

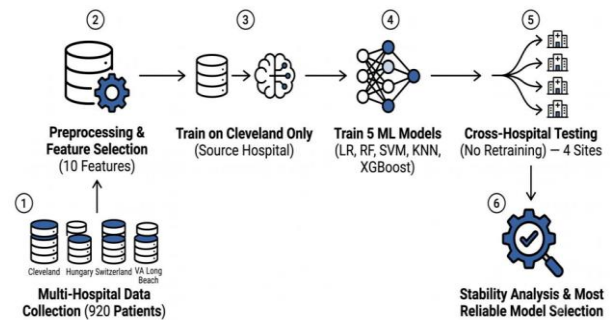


Figure 1: Cross-Hospital Generalization Architecture for Heart Disease Prediction

Figure 1 illustrates the complete workflow: multi-hospital data collection from four sites (Cleveland, Hungary, Switzerland, VA Long Beach), preprocessing and feature selection (10 common features), training exclusively on Cleveland (source hospital), training five machine learning models, cross-hospital testing on all four sites without retraining, and final stability analysis to identify the most reliable model.

III(I). Dataset Description For our experiments:

We used the well-established UCI Heart Disease repository. This dataset contains records from 920 patients drawn from four independent medical institutions: Cleveland, Hungary, Switzerland, and the VA Long Beach center. The dataset is publicly available and has been extensively utilized in predictive modeling studies.

We narrowed down the feature set to 10 variables that were consistently recorded across all four sites. These ten features are: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic results, maximum heart



rate achieved, exercise-induced angina, and ST depression (oldpeak).

We simplified the original multiclass target into a binary label—0 for no disease and 1 for disease present. Several features were excluded from the analysis because they were largely missing in the Hungary, Switzerland, and VA Long Beach datasets, and imputing them would have introduced more guesswork than signal.

III(II). Data Preparation and Split Strategy:

All data cleaning and preprocessing steps were performed using only the Cleveland partition. Missing numeric values were replaced with the median from Cleveland, and missing categorical values were replaced with the mode from Cleveland. This decision was critical to prevent information from the other three hospitals from influencing the training data. For scaling, we applied standard normalization (zero mean, unit variance). The normalization parameters were learned exclusively from the Cleveland training portion and then applied to all other datasets. This simulates a realistic scenario in which a model is developed in one site and then transported to others. The training set was formed by randomly selecting 80% of the Cleveland records. The remaining 20% of Cleveland, along with the complete datasets from Hungary, Switzerland, and VA Long Beach, were set aside as evaluation partitions.

III(III). Novelty of the Experimental Design:

The key differentiator of our study lies in the evaluation scheme. Most earlier works have combined all sites. and performed random splits, we intentionally trained classifiers on a single site. This allows us to measure population-level generalizability, which is a more demanding and practically relevant test than within-sample accuracy. Our focus is not simply on predictive power, but on consistency across varying clinical populations.

III(IV). Model Selection We selected a set of five models, each representing different learning paradigm, to ensure a comprehensive comparison of model behavior across hospital populations.

The chosen models include:

- Logistic Regression (linear classifier)
- Random Forest (bagged ensemble of trees)
- Support Vector Machine (SVM)
- K-Nearest Neighbors (distance-based lazy learner)
- XGBoost (boosted tree model)

All models were implemented using their default hyperparameters, with a fixed random seed (random_state = 42) applied throughout to ensure reproducibility. The one exception was XGBoost's learning rate, which was reduced from its default value of 0.3 to 0.1.

III(V). Evaluation Protocol:

For evaluation of all models a range of standard classification metrics were computed to provide a comprehensive assessment of predictive ability of the models. This included:

- Accuracy – the proportion of correctly classified instances out of the total predictions.
- Precision – the ratio of true positive predictions to the total positive predictions, indicating how often the model is correct when it predicts disease.
- Recall (Sensitivity) – the ratio of true positive predictions to the actual positive instances, reflecting the model's ability to correctly identify patients with heart disease.
- F1 Score – the harmonic mean of precision and recall. It provides a balanced measure of performance, making it especially useful when both false positives and false negatives carry significant consequences, as is the case in medical prediction tasks.



The F1 Score was prioritized due to its balance between false alarms and missed detections.

To assess how consistently each model performs across different hospital populations, we introduced a stability metric called "Spread." Spread is defined as the range between the maximum and minimum F1 Score achieved by a model across the four hospitals.

Lower spread values indicate more consistent performance across heterogeneous populations, which we interpret as a proxy for methodological robustness rather than a claim of clinical readiness.

V. RESULTS AND DISCUSSION

This section presents the experimental findings obtained from evaluating five machine learning models across four hospital populations. All models were trained exclusively on the Cleveland dataset and tested on the remaining three hospitals without any retraining. We report standard classification metrics, with particular emphasis on F1 Score and the stability metric (Spread) introduced in the previous section.

Overall Performance

Table 1 presents the complete set of results—Accuracy, Precision, Recall, and F1 Score—for each of the five models across all four hospitals. The Cleveland column represents performance on the held-out 20% test split from the training hospital, while the remaining columns reflect performance on entirely unseen patient populations.

Table I: Example Table Title

Model	Hospital	Accuracy	Precision	Recall	F1 Score
Data	Data B	Data C			

V. CONCLUSION

Summarize the key findings of your research and their implications for theory, practice, or policy. Highlight the significance of your contribution and suggest areas for future work. Avoid repeating sentences from the abstract.

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REFERENCES

- [1] A. A. Author and B. B. Author, "Title of paper," Journal Name, vol. 10, no. 2, pp. 100–120, Year.
- [2] C. C. Author, "Title of Book," xth ed. City, Country: Publisher, Year.
- [3] D. D. Author et al., "Conference Paper Title," in Proc. IEEE Conf., Year, pp. 1–6.